1. O/N/21/22/Q4

A washing machine is working normally with both the water heater and the motor switched on. The washing machine is connected to the mains supply by a cable.

(a) The current in the live wire in the cable is 13A.

State the size of the current in:

(i) the neutral wire in the cable

current in neutral wire = ... [1]

(ii) the earth wire in the cable.

current in earth wire = ... [1]

(b) The insulation on the mains cable is now damaged and, as the washing machine vibrates, the live wire touches the metal casing.

(i) Explain how the earth wire and the fuse together prevent any more damage.

....

....

....

....

.... [3]

(ii) Explain why it is the live wire into which the fuse is connected.

....

....

.... [2]

[Total: 7]

2. O/N/21/22/Q9

A filament lamp is connected to a 240 V alternating current (a.c.) mains supply.

(a) Describe how the output of an a.c. supply differs from the output of a direct current (d.c.) supply.

Sketch **two** voltage–time graphs in the blank space to help your explanation.

....

.... [3]

(b) The lamp is rated at 60 W and is designed to be used with a 240 V supply.

(i) Calculate the current in the lamp.

current = ... [2]

(ii) Calculate the resistance of the lamp.

resistance = ... [2]

(iii) A room is lit by five of these filament lamps connected in parallel.

State **two** advantages of connecting the lamps in parallel rather than in series.

1. ....

....

2. ....

....

[2]

(iv) The five lamps in **(b)(iii)** are lit for an average time of 5.5 hours a day for a year. Electricity costs \$0.15 / kW h.

Calculate the cost of using these lamps for a year.

cost = ... [2]

(c) A student takes the lamp in **(b)** to school and connects it in a circuit using a 1.5 V cell, an ammeter and a voltmeter. The circuit is used to determine the resistance of the lamp.

(i) In the blank space, draw the circuit diagram of the circuit used to determine the resistance of the lamp.

[2]

(ii) The value of the resistance of the filament lamp in this circuit differs greatly from the value calculated in **(b)(ii)**.

State how the resistance value in this circuit differs and explain why it differs.

....

....

.... [2]

[Total: 15]

3. O/N/20/22/Q5

The cable of a washing machine contains three separate wires. There is a fuse in one of the wires.

(a) Explain how the earth wire and the fuse work together to make the washing machine safer.

....

....

....

....

.... [3]

(b) **(i)** State the name of the wire in which the fuse is connected.

.... [1]

(ii) Explain why the fuse is connected into this wire.

....

.... [1]

(c) The cable of a hair-dryer contains only **two** wires.

(i) State the name of each of these wires.

1. ....

2. ....

[1]

(ii) Suggest why the hair-dryer does not need an earth wire.

....

.... [1]

[Total: 7]

4. M/J/20/22/Q6

The clothes iron shown in Fig. 6.1 is connected to the electrical mains.

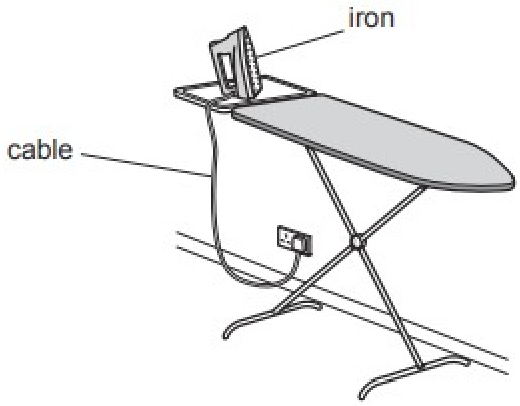

Fig. 6.1

(a) The boxes in the left column below contain some electrical hazards. The boxes in the right column contain methods of protection from these hazards.

For each hazard, draw **one** line to the appropriate method of protection.

electrical hazard	**method of protection**
worn insulation on the cable to the iron	earth wire and fuse in plug correctly connected to iron
loose live wire in the iron touches its metal case	circuit breaker correctly connected in circuit
cable becomes too hot because current is too high	visual check of cable before connecting to mains

[2]

(b) The power of the iron is 1200 W. The cost of 1 kWh of electrical energy is 20 cents (20 c).

(i) Define the *kilowatt-hour* (kWh).

....

.... [1]

(ii) The iron is on at full power for 20 minutes.

Calculate the cost of running the iron for this time.

cost = ... [2]

5. M/J/18/22/Q8

A fuse is one form of protection in an electrical circuit.

(a) State **two other** forms of protection that are included in household electrical circuits.

These may protect the consumer, the circuit or an electrical appliance.

1. ....

2. ....
[2]

(b) Fig. 8.1 shows a fusebox connected to part of a lighting circuit in a house.

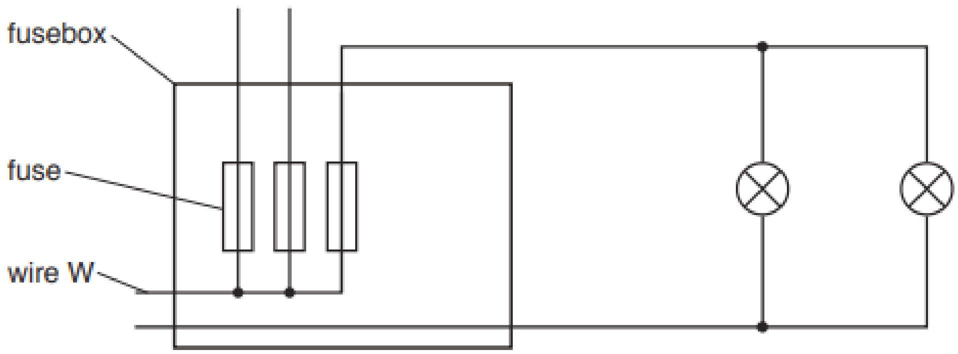

Fig. 8.1

(i) State how Fig. 8.1 shows that wire W is the live wire.

....

....[1]

(ii) On Fig. 8.1, mark with a letter s, the correct position for a switch that controls both lamps.
[1]

(iii) The rating of the fuse in the lighting circuit is 5A.

Explain what this means.

....

....

....[1]

6. M/J/17/21/Q6

Fig. 6.1 shows a 240 V a.c. mains supply connected to a television and two lamps.

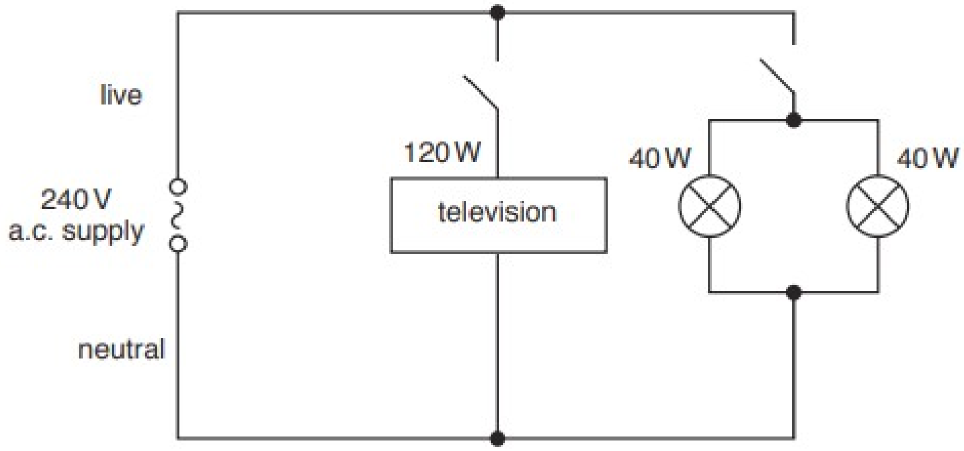

Fig. 6.1

In normal operation, the power supplied to each lamp is 40 W and the power supplied to the television is 120 W.

(a) Calculate, in normal operation,

(i) the total power supplied,

power = ...[1]

(ii) the total number of kilowatt-hours (kW h) of energy supplied to the circuit in 3.0 hours,

number of kW h = ...[2]

(iii) the current in each lamp.

current = ...[2]

(b) Explain why the switches are placed in the live wire and not in the neutral wire.

....

....

....

....[2]

7. O/N/16/21/Q7

The a.c. mains electricity system in a house consists of three wires: the live, the neutral and the earth.

(a) **(i)** State the purpose of the live wire.

....

....[1]

(ii) Explain why the neutral wire is needed.

....

....[1]

(b) Each circuit in the house includes a fuse for protection.

(i) An electrical appliance is switched on and the current in the fuse becomes greater than the rating of the fuse.

State and explain what happens.

....

....

....[2]

(ii) State and explain in which of the three wires the fuse is located.

....

....

....

....[2]

8. O/N/15/22/Q8

A spotlight in a television studio is at full brightness. The lamp inside the spotlight is powered by a 230 V supply and the current in the filament of the lamp is 27 A.

(a) Calculate the power of the lamp.

power = ... [2]

(b) The spotlight is kept switched on at full brightness.

(i) Calculate the energy transformed by the lamp in 30 minutes.

energy = ... [1]

(ii) The cost of using one kilowatt-hour (kWh) of electricity is 23 cents.

Calculate the cost of using the spotlight for 30 minutes.

cost = ... [2]

9. M/J/15/22/Q5

When a balloon is rubbed on hair, the balloon becomes negatively charged. The balloon is shown in Fig. 5.1.

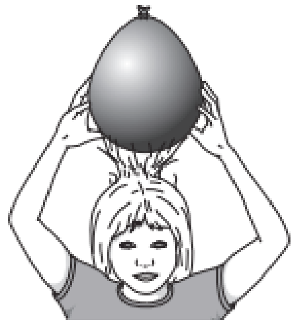

Fig. 5.1

(a) Explain how rubbing causes the balloon to become negatively charged.

....

....

.... [2]

(b) Explain why the hair is pulled towards the balloon.

....

....

....

.... [2]

(c) Explain why it is important that the balloon is made from an electrical insulator.

....

.... [1]

10. M/J/15/22/Q7

An electric hairdryer and an electric heater are connected to the mains supply, as shown in Fig. 7.1.

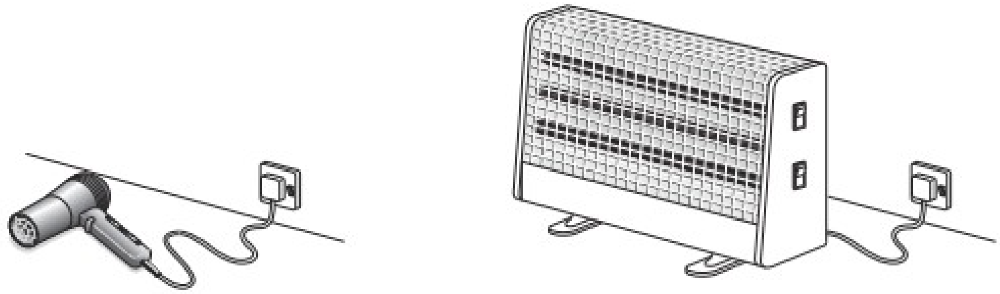

Fig. 7.1

The cable from the heater to the mains supply has a live, a neutral and an earth wire.

(a) State the purpose of the neutral wire.

....

.... [1]

(b) The live wire in the electric heater touches the outer metal case.

Explain how the earth and the fuse together protect the user from electric shock.

....

....

....

.... [2]

(c) The hairdryer does not have an earth wire. Explain why this hairdryer is still safe to use.

....

.... [1]

(d) In some modern homes, circuit breakers are used instead of fuses.

Suggest one advantage of using a circuit breaker rather than a fuse.

....

.... [1]

11. O/N/13/21/Q7

A 9.6 kW electric shower is powered by a 240 V mains supply.

(a) (i) Calculate the electric current in the heating element of the shower.

current = ... [3]

(ii) Suggest an appropriate rating for the circuit breaker in the circuit for the shower.

.... [1]

(b) A family uses the shower for a total time of 25 minutes every day. The cost of 1 kW h of electrical energy is 21 cents.

Calculate the daily cost of using the shower.

cost = ... [2]

12. M/J/12/21/Q9

Fig. 9.1a shows a room heater. Fig. 9.1b is a diagram of the electric circuit of the heater.

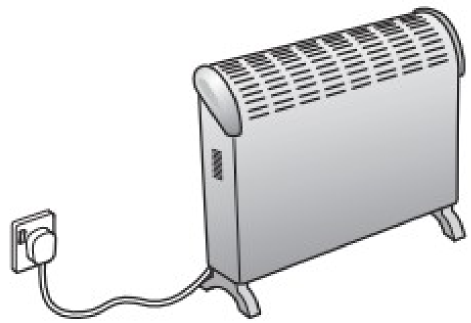

Fig. 9.1a

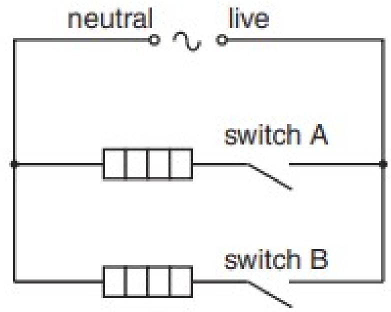

Fig. 9.1b

The fuse has not been drawn on the circuit diagram in Fig. 9.1b.

(a) (i) On Fig. 9.1b, draw the symbol for a fuse in the correct position. [2]

(ii) State the part of the room heater to which the earth wire is connected.

....[1]

(iii) The earth wire reduces the chance of an electric shock if a fault develops in the room heater.

1. State one fault that causes an electric shock when a person uses the room heater without an earth connection.

....

....[1]

2. Explain how using an earth connection prevents an electric shock.

....

....

....[2]

(b) **(i)** This type of room heater is very efficient. Explain what this means.

....

....[1]

(ii) The room heater is a convector heater. Describe and explain how thermal energy (heat) passes around a room by convection.

....

....

....

....

....[3]

(c) Fig. 9.2 shows the power output of the room heater when each switch is closed.

	power / W
switch A only closed	600
switch B only closed	
both switches closed	2100

Fig. 9.2

(i) Determine the power output of the room heater when only switch B is closed.

power output = ... [1]

(ii) The room heater is used with both switches closed for 2.5 hours.

Calculate the energy output of the room heater

1. in kilowatt-hours,

energy = ... kW h [2]

2. in joules.

energy = ... J [2]

13. O/N/11/21/Q10

Fig. 10.1 is the circuit diagram of a fan oven that works from a 230 V a.c. mains supply.

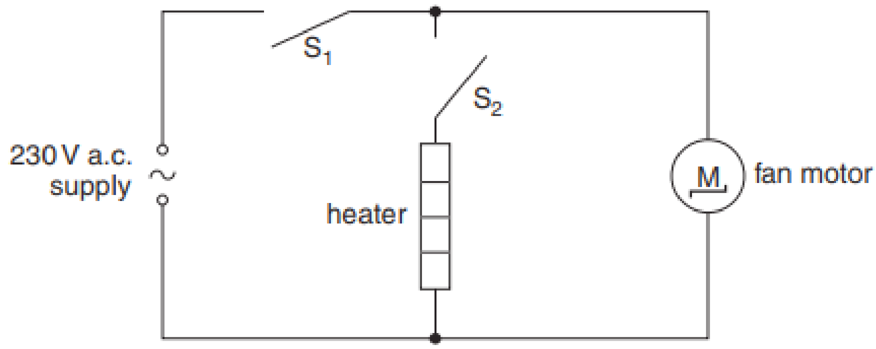

Fig. 10.1

The oven contains a 3500 W electric heater, an electric fan operated by a 200 W fan motor and two switches S_1 and S_2.

(a) (i) When switches S_1 and S_2 are open, both the heater and the fan motor are off, as shown in Fig. 10.2.

S_1	S_2	heater	fan motor
open	open	off	off
		off	on
		on	on

Fig. 10.2

Complete Fig. 10.2 by writing "open" or "closed" in the boxes. [2]

(ii) Explain how the circuit represented by Fig. 10.1 ensures that the heater cannot be switched on when the fan motor is off.

....

....

....[2]

(iii) Suggest one reason why the circuit is designed so that the heater cannot be switched on when the fan motor is off.

....

....[1]

(b) **(i)** The heater and the fan motor are both switched on. Calculate the current supplied to the oven.

current = ...[2]

(ii) **1.** Suggest a suitable rating for the fuse to be used with this oven.

....[1]

2. The fuse is placed in one of the three wires in the mains lead. State the wire in which it is placed.

....[1]

(iii) Explain how earthing the metal case of the oven makes it safer.

....

....

....

....[3]

(c) The cost of 1 kW h of electrical energy is 35 cents. Calculate the cost of leaving just the fan motor switched on for 12 hours.

cost = ...[3]

14. M/J/11/22/Q6

The cable from the mains plug to a washing machine contains a live wire, a neutral wire and an earth wire. The earth wire is connected to the metal case of the washing machine.

(a) Explain how connecting the earth wire to the metal case makes the washing machine safer.

....

....

....[2]

(b) When in use, the average input power to the washing machine is 500 W.

Calculate the number of kW h of energy used by the washing machine in 45 minutes of use.

number of kW h = ...[2]

15. O/N/09/Q6

A microwave oven is rated at 650 W and is connected to a 230 V mains supply.

(a) **(i)** Calculate the current from the supply when the microwave oven is switched on.

current = ... [2]

(ii) Suggest a rating of the fuse for use with this oven.

fuse rating = ... [1]

(b) The insulation of the mains cable has worn away. The live wire touches the outer metal casing of the microwave oven.

(i) Explain the hazard that results if the outer metal casing is not earthed.

....

....

.... [2]

(ii) Explain how connecting the earth wire to the outer casing and using a fuse of a suitable rating removes this hazard.

....

....

....

.... [2]

16. M/J/06/Q6

Fig. 6.1 shows a mains extension lead. The six sockets allow several electrical appliances to be connected to the mains supply through one cable.

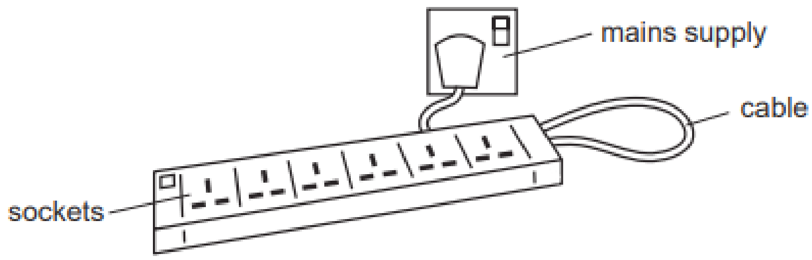

Fig. 6.1

(a) The cable connects the sockets to the mains supply.

The cable contains three wires: live, neutral and earth. State what is meant by

(i) *live*,

....

.... [1]

(ii) *neutral*,

....

.... [1]

(iii) *earth*.

....

.... [1]

(b) Six powerful lamps are plugged into the sockets and switched on, one by one.

(i) State what happens in the cable as the lamps are switched on, one by one.

....

.... [1]

(ii) Describe why it can be dangerous when a fuse of the wrong value is used in the plug.

....

....

.... [1]

(c) Explain why your hands should be dry when you put a plug into a socket.

....

.... [1]

17. M/J/05/Q6

The table gives information about two household appliances.

appliance	mains supply voltage / V	current through appliance / A	power / W	power / kW	time used per day / h	energy used per day / kW h
television	240	1.20	288	0.288	2.50	0.720
water heater	240	12.6			0.50	

(a) Write the missing values in the empty spaces in the table. [3]

(b) Why is more power needed for the water heater than for the television?

....

....

.... [1]

(c) The water heater is connected to the mains supply. Explain why using a 3 A fuse would **not** be suitable.

....

....

....

....

....

.... [2]